

Laws of Motion

Engineering Physics topic for demo testing

Newton's First Law:

A body remains at rest or in uniform motion unless an external force acts on it.

Newton's Second Law:

Force equals rate of change of momentum.

For constant mass, $F = ma$.

Newton's Third Law:

For every action, there is an equal and opposite reaction.

Inertia and Momentum:

Inertia is the tendency to resist change in motion.

Momentum $p = mv$.

Examples:

Passenger falls backward when a bus starts suddenly.

Recoil of a gun shows action and reaction.

Pushing a wall does not move it because forces balance.